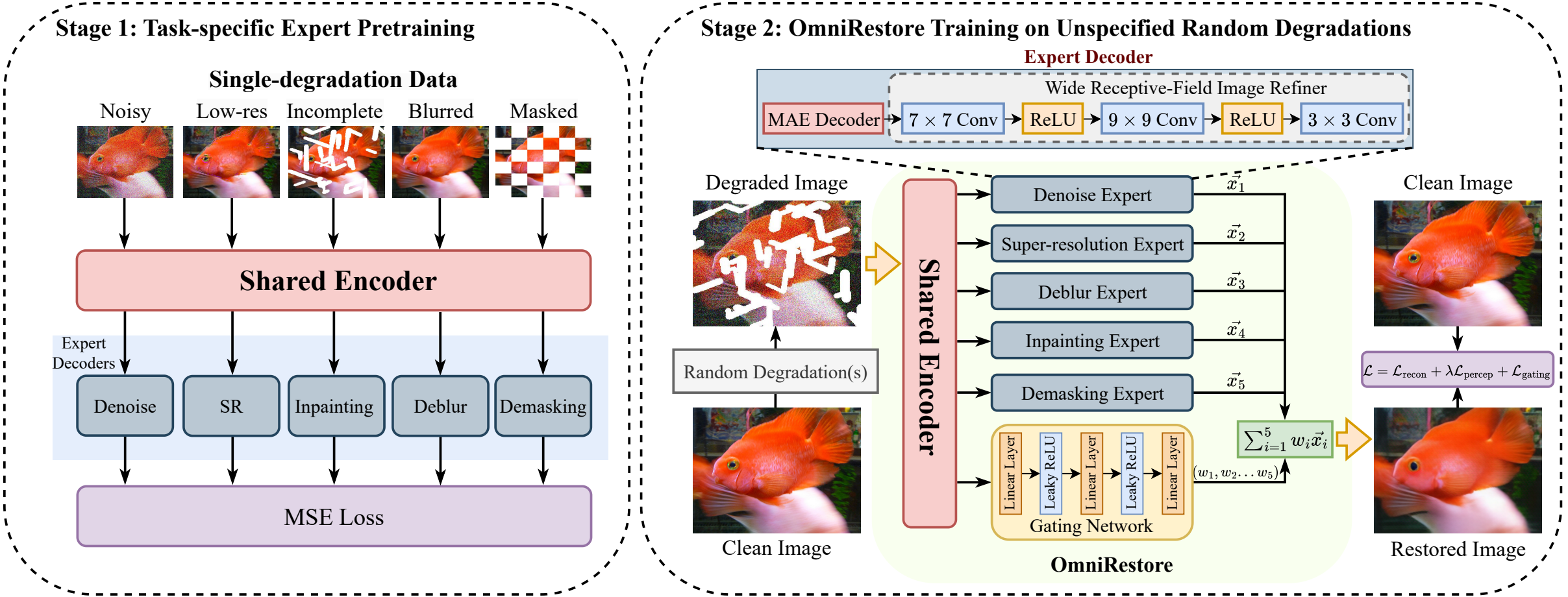
Stage 1: Task-specific Expert Pretraining
Single-degradation Data
Noisy
Low-res
Incomplete
Blurred
Masked
Shared Encoder
Expert Decoders
Denoise
SR
Inpainting
Deblur
Demasking
MSE Loss
Stage 2: OmniRestore Training on Unspecified Random Degradations
Expert Decoder
Wide Receptive-Field Image Refiner
MAE Decoder
7 × 7 Conv
ReLU
9 × 9 Conv
ReLU
3 × 3 Conv
Degraded Image
Random Degradation(s)
Clean Image
Shared Encoder
Denoise Expert
Super-resolution Expert
Deblur Expert
Inpainting Expert
Demasking Expert
$\vec{x}_1$
$\vec{x}_2$
$\vec{x}_3$
$\vec{x}_4$
$\vec{x}_5$
Linear Layer
Leaky ReLU
Linear Layer
Leaky ReLU
Linear Layer
Gating Network
$(w_1, w_2 \ldots w_5)$
$\sum_{i=1}^{5} w_i \vec{x}_i$
OmniRestore
Clean Image
$\mathcal{L} = \mathcal{L}_{\text{recon}} + \lambda \mathcal{L}_{\text{percep}} + \mathcal{L}_{\text{gating}}$
Restored Image